Jayant Varma

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Education

York University, MSc. in Computer Science September 2025 – present

- Supervisor: Dr. Matthew Kyan
- Thesis area: 3D computer vision, scene representations, representation learning
- Coursework: Big data systems (A+), Mining software engineering repositories(A), Probabilistic models and Machine Learning(A+), Digital Image Processing: Theory and Algorithms(A), Data analytics and visualization (A).

York University, BSc. Honors with Specialization in Computer Science January 2021 – June 2025

- Overall GPA: 7.14/9 (79.34% or 3.3/4)
- Final year Coursework: Research Project course in Computer Vision, Signals and Systems, Computer Vision, Machine learning, Mathematics of Cryptography, Distributed systems, Advanced data structures, Compilers and interpreters, Computational Physics, Advanced topics in Computer Vision[upper year grad course, audited]

Publications

Markus D. Solbach, Mohit Goyal, Sakar Khattar, **Jayant Varma**, Bjorn Vlaskamp, John Tsotsos, Konstantine Tsotsos, *"Passthrough Rigidity: The Behavioral and Visuomotor Costs of Mediated Perception"*, ISMAR Conference, 2026

Markus D. Solbach, Mohit Goyal, Sakar Khattar, **Jayant Varma**, Bjorn Vlaskamp, John Tsotsos, Konstantine Tsotsos, *"Visual and Motor Behaviour for Object Manipulation in a Dynamic Environment"*, Journal of Vision, 2026

Experience

Teaching Assistant, Dept. of Electrical Engineering and Computer Science, York University - Toronto, ON, Canada September 2025 – present

- Held tutorials, labs, lab tests, and exams for Computer Organization, Data structures course
- Teaching quality hands on expertise with low level programming, RISC-V assembly, data structures and algorithms

Research Assistant, Tsotsos Laboratory for Active and Attentive Vision, Dept. of Electrical Engineering and Computer Science, York University - Toronto, ON, Canada April 2025 – August 2025

- Provided experimental, coding and data analysis support for human investigations into natural and digitally assisted vision
- The project was made in collaboration with the Google AR/VR group
- Co-author of two papers at VSS (accepted) and ISMAR (accepted)

Computer Vision and Machine Learning Researcher, Elder Laboratory – Toronto, ON, Canada January 2024 – April 2025

- Research project course, EECS 4080
- Did literature survey of:
  - Autonomous driving datasets of indoor and outdoor scenes (real and synthetic)
  - Lightweight Neural net methods for instance and panoptic segmentation for autonomous driving
- Fine-tuned and trained YOLO11n on the SceneParse150 and the COCO-Panoptic datasets to perform instance segmentation for navigation of the AirChair wheelchair
- Post-processing using Robot Operating System (ROS), and C++ to optimize the code to perform in real-time.
- Video presentation link

Volunteer researcher, Allan I. Carswell Observatory, York University – Toronto, ON September 2024 – April 2025

- Rediscovered Kelt-16 b exoplanet using the transit method, thus verifying it's existence and reproducing the results of the original discovery paper, supervised by Professor Sarah Rugheimer and Joshua Parsons

  - Created a Python data analysis pipeline for processing images using NumPy, AstroPy, Matplotlib, SciPy
  - Image analysis is heavy on computer's RAM, I made a clever, optimized algorithm that works with as little resources as necessary
  - Did technical writing in LaTeX
- Help conduct, and do my own astronomy research at Allan I. Carswell Observatory

Computer Vision and Machine Learning Researcher, Universität Augsburg – Augsburg, Bavaria, Germany June 2024 – September 2024

- Prestigious DAAD RISE-Germany scholarship holder, co-funded by DAAD and Mitacs as one of top applicants from Canada
- Worked with Julian Lorenz at Dr. Rainer Lienhart's lab to create a synthetic dataset of indoor scenes that also features humanoids. Scenes and humanoids were generated using diffusion models, and key annotations such as scene graphs, 3D point clouds, object segmentations, and more (link to list of annotations) were added to show human-object, object-object, and human-human relations
- Implemented key parts of the pipeline bridging the diffusion model output to JSON to Unity
- Programmed in C# and worked in Unity to add objects in the generated scenes and humanoids, and also added scene graph annotations
- Presented paper on diffusion model that generates scenes, and read about inpainting, scene graphs, scene and humanoid synthesis stable diffusion models

Class representative, Bethune College, York University – Toronto, ON September – December 2024

- Held and organized study sessions for PHYS 3070: Planets and planetary systems
- Taught Python, NumPy, and LaTeX
- Strong interpersonal skills

Class representative, Bethune College, York University – Toronto, ON January 2023 – April 2023

- Held and organized study sessions for Math 1025: Applied Linear Algebra
- Helped many new students prepare better for exams and answered their questions, solved questions with them
- Strong interpersonal skills

Clubs and group activities

Computer Education Reading Group (CERG) Lassonde School of Engineering October 2022 – present

- First student member of the group where we discuss papers and ways to teach Computer Science to improve Computer Science pedagogy at Lassonde

Club Inifinity, York University May 2021 – Present

- Active member of the math club's reading group from Jan.2022 to Aug.2022 where we read *Gödel, Escher, Bach* by Douglas Hofstader over the course of Winter and Summer of 2022

Awards

Graduate student fellowship (2025): $78,990

DAAD RISE Germany Scholarship funded by Mitacs Globalink (2024): $6,000

York University continuing student scholarship(2023): $600
Two time EECS 3101 (Algorithms) Learning Excellence award Winter 2023
EECS 2011 (Data structures) Learning Excellence award Fall 2022
York University International Scholarship of Merit(2021): $4,000

Technical Skills

Programming Languages: C, Java, Python, Go, C#, JavaScript, Matlab

Others: Pandas, pyxdf, Git/Github, Unity, PyTorch, Kornia, NumPy, JAX, OpenCV, Kubernetes, Azure, Docker, Redis, LaTeX, HTML/CSS, Linux, MacOS, Windows, MSOffice